In [1]:
```
import seaborn as sns
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```
xyz = pd.read_csv('HRDataset_v14.csv')
xyz.head()
```

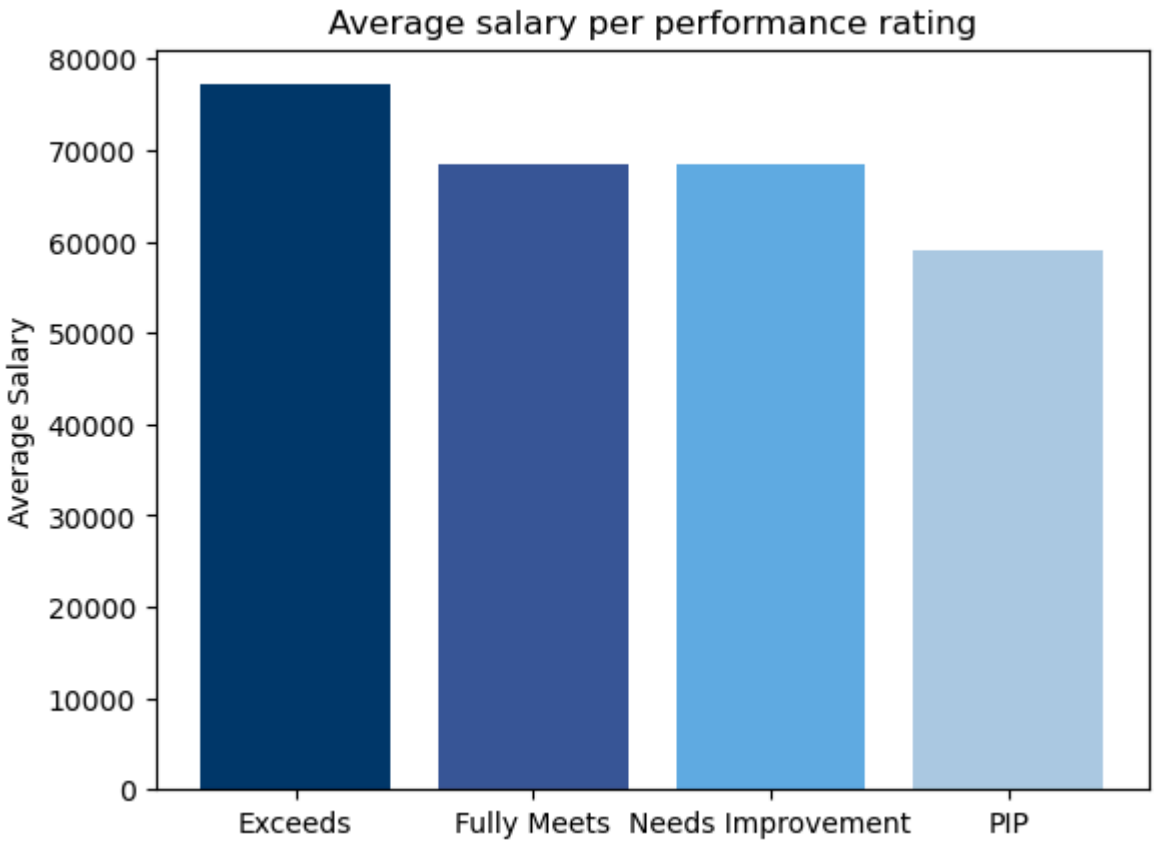
Out[2]:

	EmpID	Salary	PositionID	Position	State	Zip	DOB	Sex	MaritalDesc	CitizenDesc	...	ManagerName	ManagerID	Recruit
0	10026	62506	19	Production Technician I	MA	1960	07/10/1983	M	Single	US Citizen	...	Michael Albert	22	
1	10084	104437	27	Sr. DBA	MA	2148	05/05/1975	M	Married	US Citizen	...	Simon Roup	4	
2	10196	64955	20	Production Technician II	MA	1810	09/19/88	F	Married	US Citizen	...	Kissy Sullivan	20	
3	10088	64991	19	Production Technician I	MA	1886	09/27/88	F	Married	US Citizen	...	Elijiah Gray	16	
4	10069	50825	19	Production Technician I	MA	2169	09/08/1989	F	Divorced	US Citizen	...	Webster Butler	39	G

5 rows × 26 columns

In [191…
```
salary_perf = xyz.groupby('PerformanceScore', as_index=False)['Salary'].mean()
colors = ['#03396c', '#3b5998', '#63ace5', '#adcbe3', '#e3f0ff','#fad9c1']
plt.bar(salary_perf['PerformanceScore'],salary_perf['Salary'],color= colors)
plt.title('Average salary per performance rating')
plt.ylabel('Average Salary')
```

Out[191…
```
Text(0, 0.5, 'Average Salary')
```



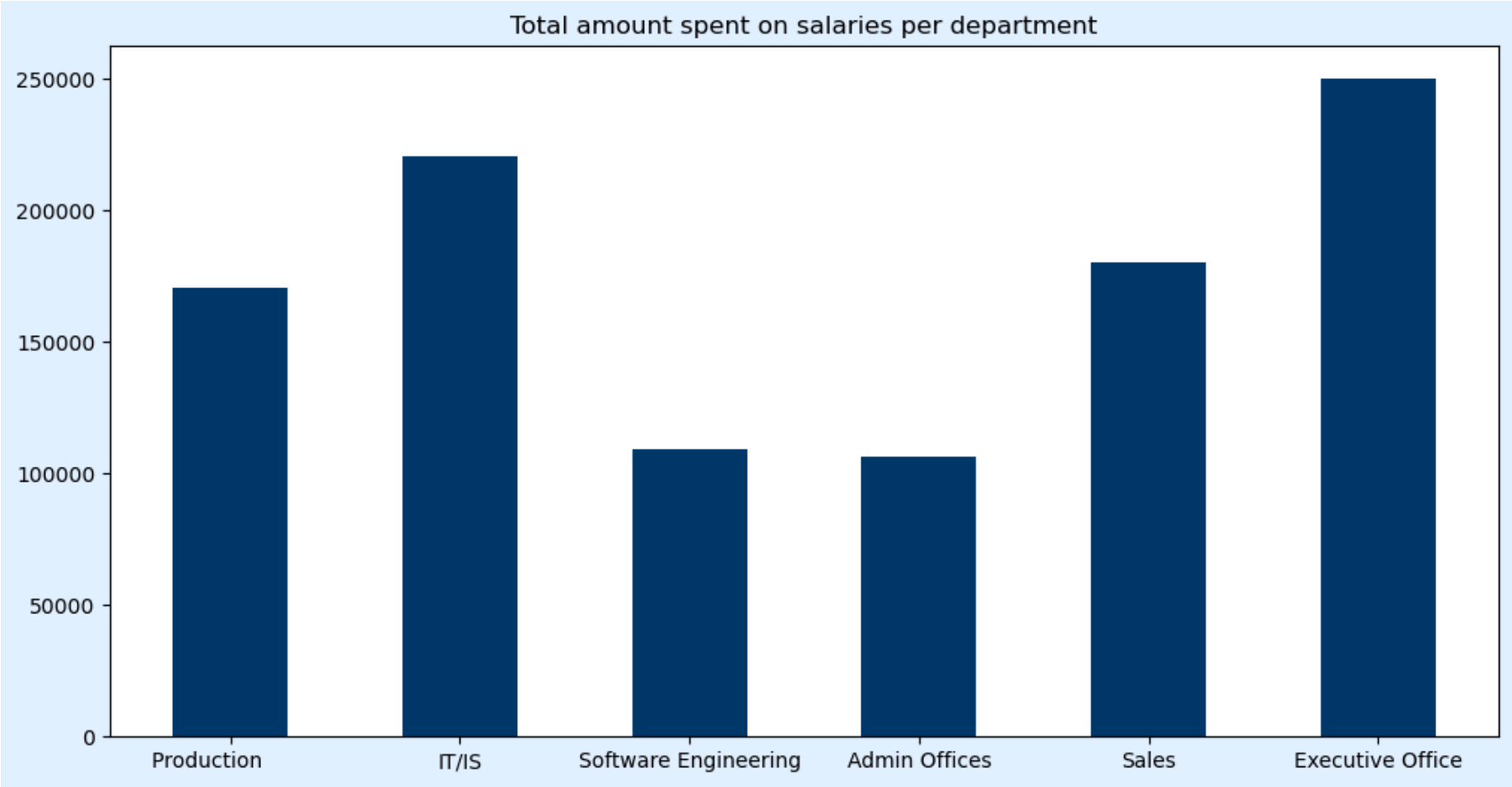
From the bar graph above we can see that the employees with a high performance rating generally earn more. The only abnormarly is that on average employees with a performance rating of "fully meets" and "Needs Improvement" earn the same eventhough they don't perform the same.

In [183…

```
#Wage to department
depart_sal = plt.figure(figsize=(12,6))
depart_sal.set_facecolor('#e3f0ff')
plt.bar(xyz['Department'],xyz['Salary'],color = '#03396c', width= 0.5)
plt.title('Total amount spent on salaries per department')
#plt.xticks(rotation = 45)
```

Out[183…

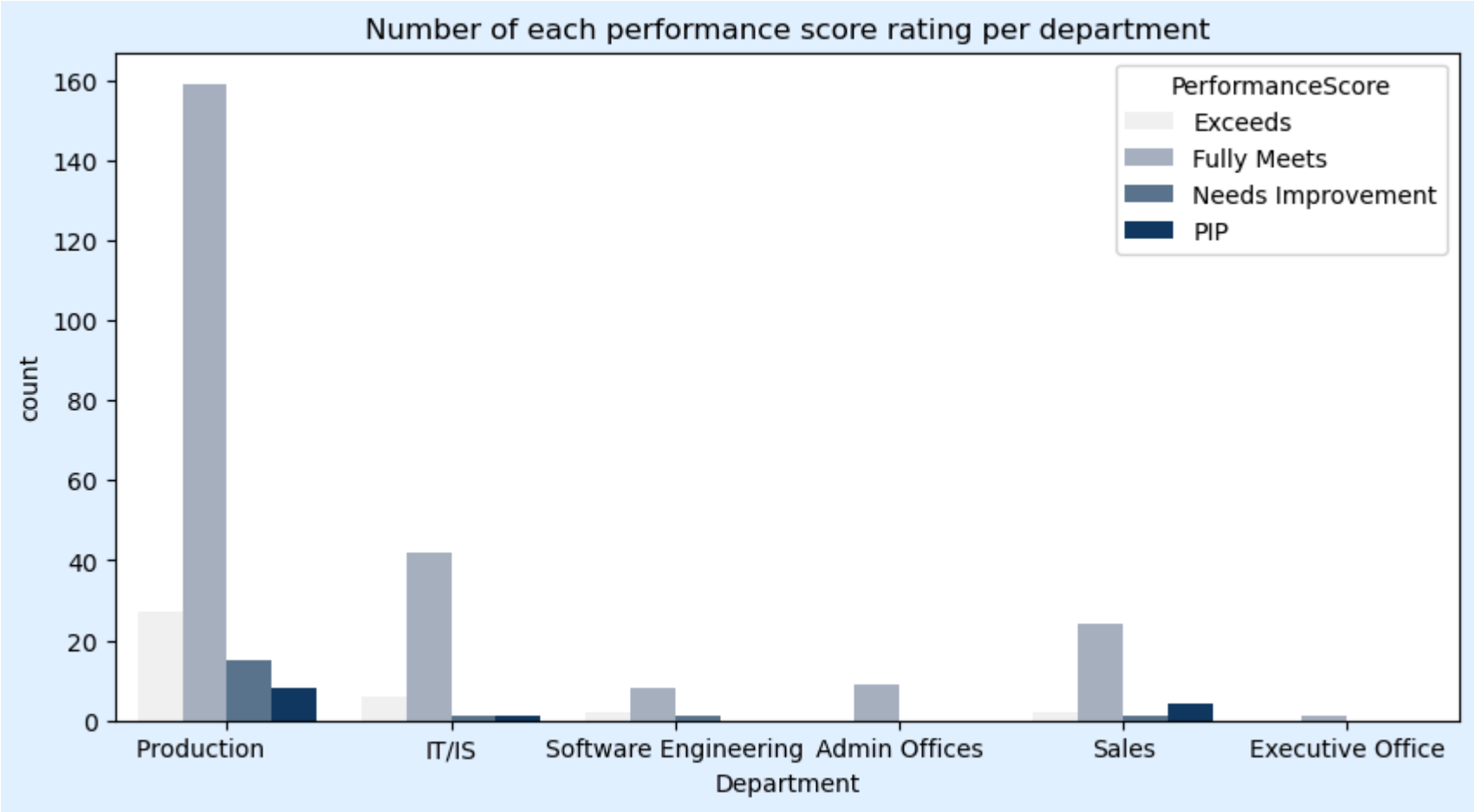
```
Text(0.5, 1.0, 'Total amount spent on salaries per department')
```



It is clear that the Executive office department spends the most on salaries followed closely by the IT/IS department. The admin offices and software engineering office spent the least on salaries.

In [192…

```
count_perf = plt.figure(figsize=(10,5))
count_perf.set_facecolor('#e3f0ff')
sns.countplot(x = 'Department', data = xyz, hue = 'PerformanceScore',color='#03396c')
plt.title('Number of each performance score rating per department')
Depart_perf = pd.DataFrame([xyz['Department'],xyz['PerformanceScore']])
```

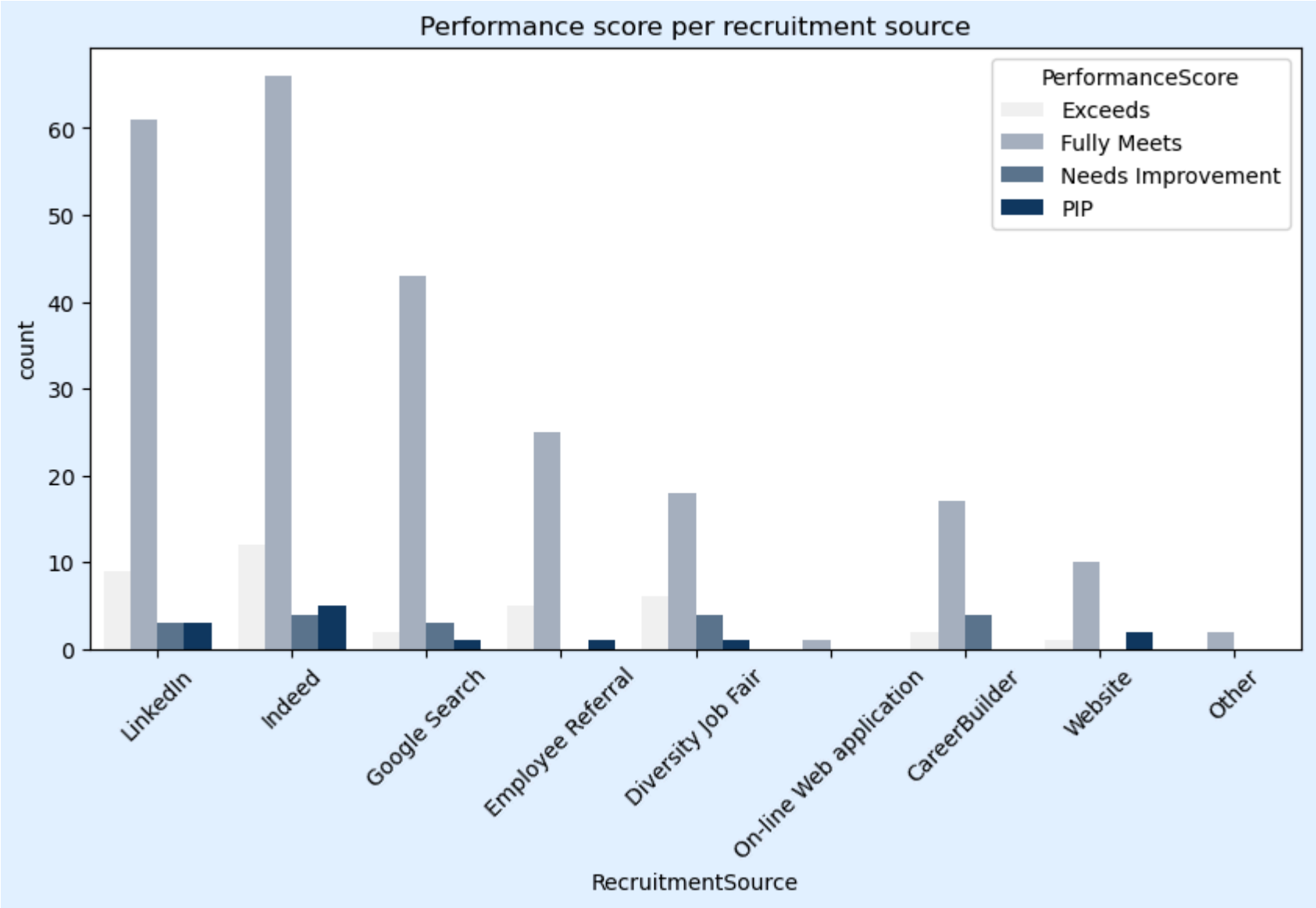


Despite the executive office being the highest spender, they have the lowest performance score ratings.Despite being just 4th in salary expenditure the production department tops the chart for performance ratings by a huge margin. Something that isn't suprising is that the admin's office which had a the lowest expendtiture on salaries got the second lowest performance rating with just a handful of employees who fully meets the standards and none that exceed it.

In [185…

```
recruit_perf = plt.figure(figsize=(10,5))
recruit_perf.set_facecolor('#e3f0ff')
sns.countplot(x = 'RecruitmentSource', data = xyz, hue = 'PerformanceScore', color = '#03396c')
plt.xticks(rotation = 45)
plt.title('Performance score per recruitment source')
```

Out[185…
```
Text(0.5, 1.0, 'Performance score per recruitment source')
```



From the graph we can see that the best performance rating came from employees who were recruited through indeed followed closely by those from linkeldn. The least productive method of recruitment was a diversity job fair and other methods.

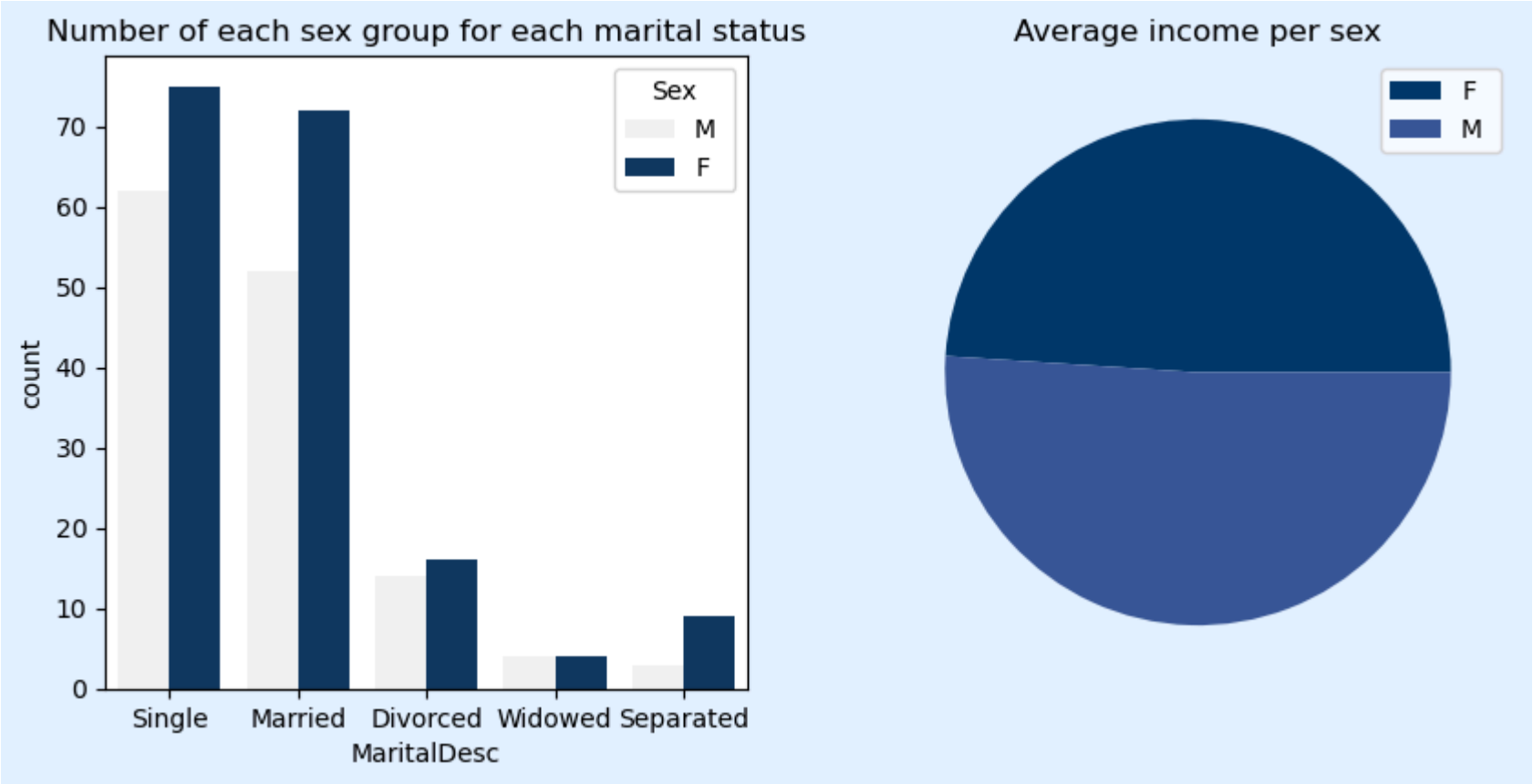
In [186…

```
base = plt.figure(figsize=(10,4.5))
base.set_facecolor('#e3f0ff')
plt.subplot(1,2,1)
sns.countplot(x = 'MaritalDesc', data = xyz, hue='Sex',color ='#03396c')
plt.title('Number of each sex group for each marital status')

plt.subplot(1,2,2)
Sex_salary = xyz.groupby('Sex', as_index=False)['Salary'].mean()
plt.pie(Sex_salary['Salary'], colors = colors)
plt.legend(Sex_salary['Sex'])
plt.title('Average income per sex')
```

Out[186…

```
Text(0.5, 1.0, 'Average income per sex')
```



The first figure above showing the number of each sex group for each marital status generally shows that there are more females in the firm than males and most of them are either single or married. The female population generally dominate in all martital categories except the widowed category which is equal to that of males. From the second chart showing the average income between male and female population we can see that despite the male population being lower they make more money on average, suggesting some form of inequality.

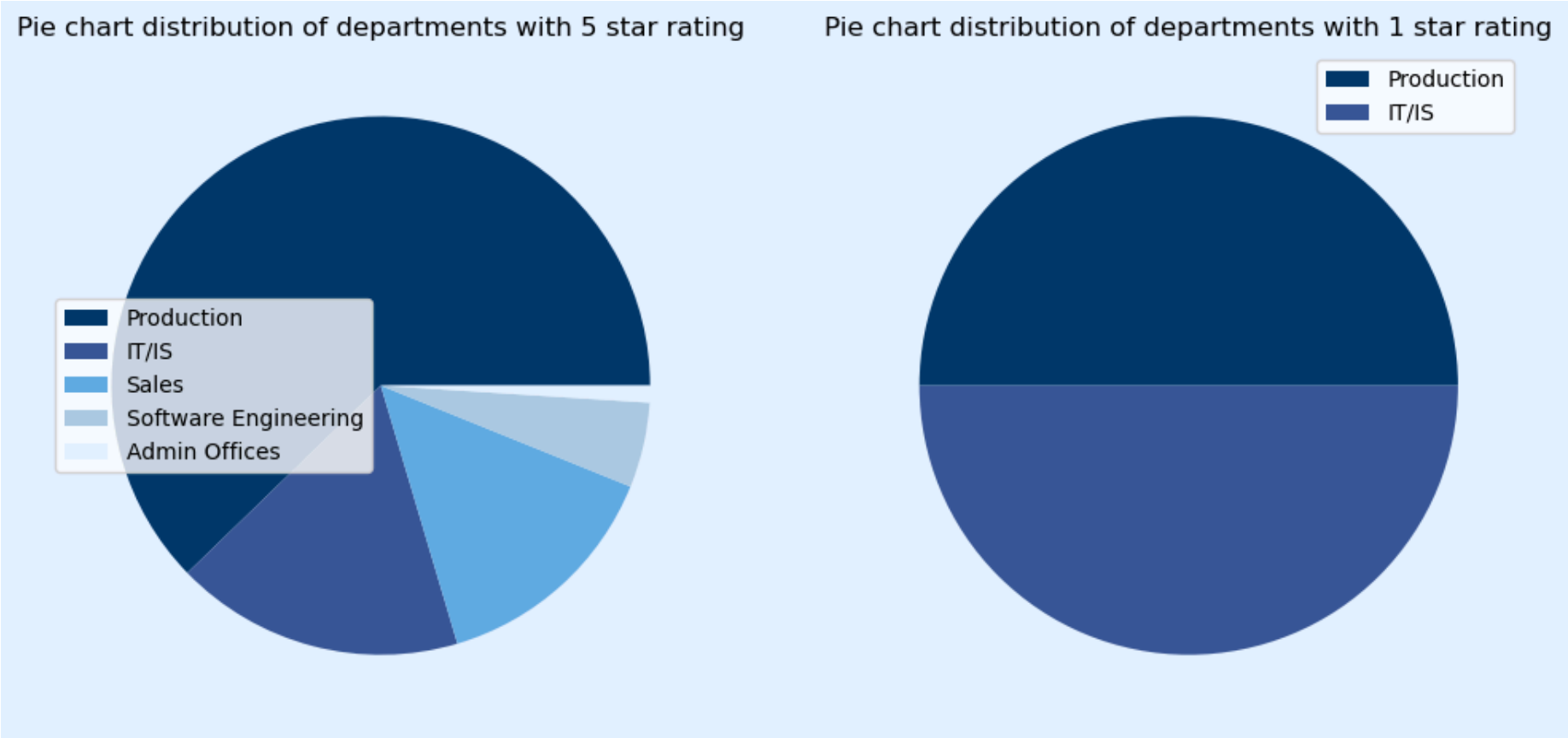
In [187…

```
#Most satisfied departments
dep_sat_5 = xyz.loc[xyz['EmpSatisfaction'] == 5,['Department']]
dep_sat_1 = xyz.loc[xyz['EmpSatisfaction'] == 1,['Department']]
base_2 = plt.figure(figsize=(12,12))
base_2.set_facecolor('#e3f0ff')
plt.subplot(1,2,1)
plot = dep_sat_5['Department'].value_counts()
colors = ['#03396c', '#3b5998', '#63ace5', '#adcbe3', '#e3f0ff']
plt.pie(plot, colors=colors)
plt.legend(['Production','IT/IS','Sales','Software Engineering','Admin Offices'])
plt.title('Pie chart distribution of departments with 5 star rating',loc='center')

plt.subplot(1,2,2)
plot_1 = dep_sat_1['Department'].value_counts()
plt.pie(plot_1, colors=colors)
plt.title('Pie chart distribution of departments with 1 star rating',loc='center')
plt.legend(['Production','IT/IS','Sales','Software Engineering','Admin Offices'])
```

Out[187…

```
<matplotlib.legend.Legend at 0x183ae4e4550>
```

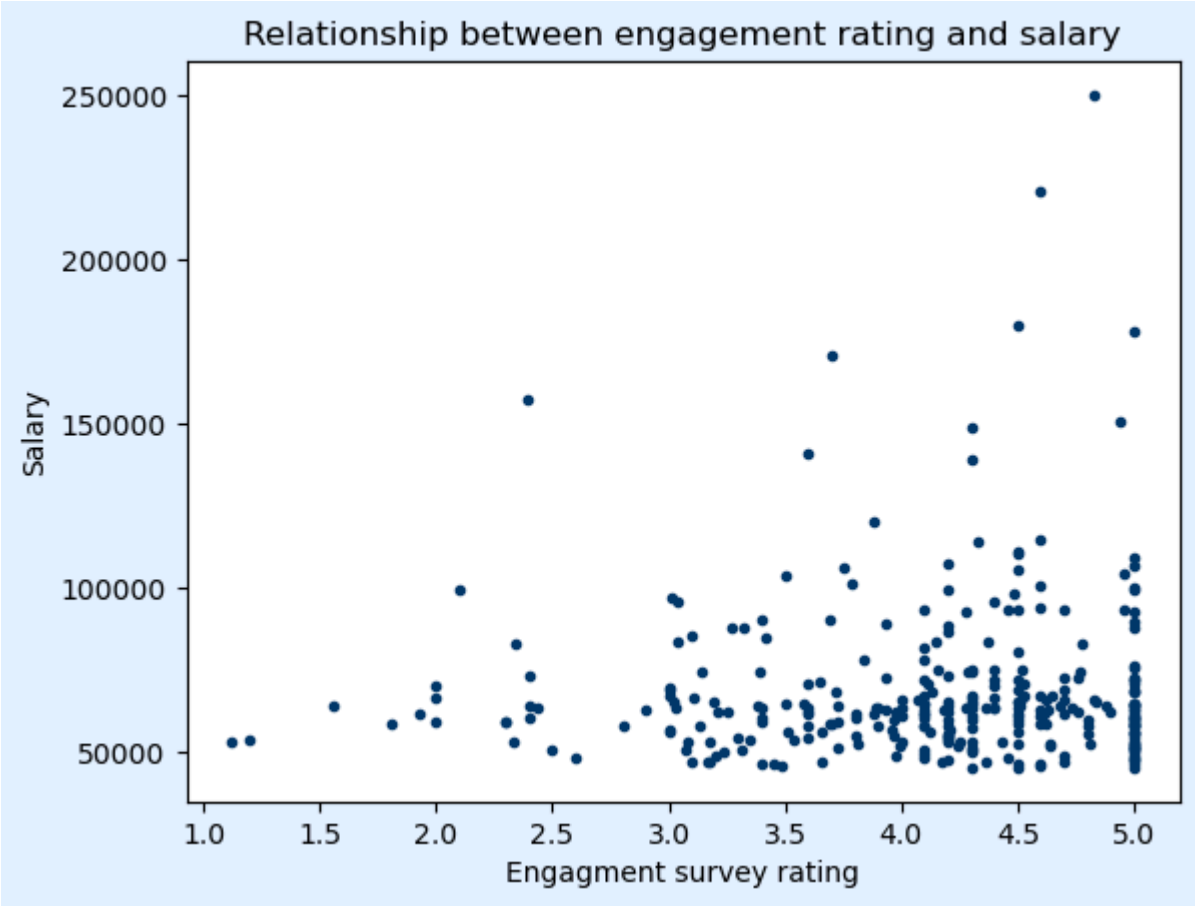


From the chart we can see that the produtcion department got both the most 5 star ratings and the 1 star rating which is tied with that of the IT/IS department. Suggesting that the department's rating varies greatly amongst employees. It also shows that the second most highly rated department which is the IT/IS, also follows this same pattern. The department with the smallest number of 5 star ratings is the admin office which on the previous graph had the lowest salaries. Which suggest some relationship between the two factors.

In [190…

```
fig, ax = plt.subplots()
fig.set_facecolor('#e3f0ff')
ax.scatter(xyz['EngagementSurvey'],xyz['Salary'], color ='#03396c', s = 9)
plt.title('Relationship between engagement rating and salary')
ax.set_xlabel('Engagment survey rating')
ax.set_ylabel('Salary')
```

Out[190…
```
Text(0, 0.5, 'Salary')
```



From the graph we can see that most of the survey ratings are found between the income range of 5000 to 150000, and that majority of these ratings are between 3 and 5. We can also observe that just one of the ratings for employees with an income range of above 150000 is below 3. Unlike that of employees with incomes less than 150000 which got multiple points below a rating of 3.